

Question Paper

Exam Date & Time: 26-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025

Number Theory and Cryptography [IT_3130]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) a. Compute the number of integers between 000 and 999 (inclusive) which satisfy $n \equiv 7 \pmod{11}, n \equiv 3 \pmod{13}$? (5)
A) b. Construct the Additive and multiplicative inverse modulo 8 Caley table.

Course Outcome: "CO1" / Cognitive Level: "Application"

- B) a. Apply the concept of Euclid's Algorithm to find the GCD of 385 and 792, and hence express it in the form of $385x + 792y$. (5)
b. Illustration CIA model.

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 2) Compute the remainder when 23^{130} is divided by 25. (5)

- A) Course Outcome: "CO2" / Cognitive Level: "Application"

- B) Solve the following system of congruences using the Chinese remainder theorem. Show your steps and reduce your answer to an element in Z_{746130} . (5)

$$\begin{aligned}x &\equiv 2 \pmod{11}, \\x &\equiv 3 \pmod{14}, \\x &\equiv 4 \pmod{15}, \\x &\equiv 5 \pmod{17}, \\x &\equiv 6 \pmod{19}.\end{aligned}$$

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 3) Let $(R, +, *)$ be an algebraic structure that satisfies all the ring axioms except commutativity of addition. Show that if R has a unity, then R is a ring. (5)

- A) Course Outcome: "CO2" / Cognitive Level: "Application"

- B) a. Using CRT, solve for x for the following (5)
 $x \equiv 2 \pmod{3}$
 $x \equiv 3 \pmod{5}$
 $x \equiv 2 \pmod{7}$
b. Find all the primitive roots of 25.

Course Outcome: "CO2" / Cognitive Level: "Analysis"

4) Using this Playfair matrix (5)

A)

J/K	C	D	E	F
U	N	P	Q	S
Z	V	W	X	Y
R	A	L	G	O
B	I	T	H	M

Encrypt this message:

I only regret that I have but one life to give for my country.

Course Outcome: "CO3" / Cognitive Level: "Application"

B) Consider 0-based alphabet indexing. You are given a plain text "COMPUTE" which is supposed to be converted as follows : (each letter's index value + 12) mod 26 to form a new plain text. Encrypt this new plain text with a 3 * 3 key matrix as shown below (5)

$$\text{Key} = \begin{matrix} \text{ASCII of } z & \text{ASCII of } C & \text{ASCII of } A \\ \text{ASCII of } z & \text{ASCII of } C & \text{ASCII of } A \end{matrix} \begin{matrix} \text{pie}(25) & \text{ASCII of } A & 5 \\ 9147 \bmod 26 & 65 & \\ \text{pie}(70) & 33 & \end{matrix}$$

Compute the ciphertext.

Course Outcome: "CO3" / Cognitive Level: "Analysis"

5) Suppose the DES F function mapped every 32-bit input R, regardless of the value of the input K, to; (5)

A)

a. 32-bit string of zero
b. R

Then
1. What function would DES then compute?
2. What would the decryption look like?
Justify your answer with illustration.

Course Outcome: "CO4" / Cognitive Level: "Application"

B) You are given the following plaintext block (in hexadecimal) (5)

P = [32, 43, F6, A8, 88, 5A, 30, 8D, 31, 31, 98, A2, E0, 37, 07, 34] and the 16-byte cipher key
K = [2B, 28, AB, 09, 7E, AE, F7, CF, 15, D2, 15, 4F, 16, A6, 88, 3C]

Perform the first round of AES encryption. (In mix columns multiply with the below matrix)

$$M = \begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix}$$

Course Outcome: "CO4" / Cognitive Level: "Application"

-----End-----

Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	IT_3130: Number Theory and Cryptography

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	20
3	3	CO 3	10
4	4	CO 4	10

* **NA** - CO Not Selected

Event Name	Subject
------------	---------

058-MIT BLR-MAKEUP EXAM-2025-2026

IT_3130: Number Theory and Cryptography

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	6	30(60%)
4	Analysis	4	20(40%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	10	50 (100%)